DREAM TEAM / SERVICES

# MASTER SOFTWARE REQUIREMENTS SPECIFICATION

## DREAM TEAM AI DEVELOPMENT OS

An autonomous, mastery-based employee development and competition platform

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| **Status** | Baseline specification - ready for stakeholder review |
| **Prepared for** | DREAM TEAM SERVICES |
| **Classification** | Confidential - internal product and engineering use |

### DOCUMENT PURPOSE

This is the single master pre-development specification for scope, behaviours, AI operating rules, data requirements, security, acceptance criteria, and verification.

*Prepared as a requirements baseline. Numerical thresholds and role weightings marked "proposed defaults" must be calibrated during the pilot.*

# Document Control

| Field | Value |
|---|---|
| **Document title** | Master Software Requirements Specification (MSRS) |
| **Product** | DREAM TEAM AI Development OS / Intelligence Arena |
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| **Owner** | DREAM TEAM SERVICES |
| **Primary audience** | Product owner, engineering, AI implementation, QA, security, operations |
| **Normative language** | MUST = required; SHOULD = recommended; MAY = optional |

## Revision History

| Version | Date | Summary | Status |
|---|---|---|---|
| **1.0** | 16 Oct 2026 | Initial master specification from the agreed DREAM TEAM AI Development OS requirements. | Baseline |

## Approval / Review Record

| Role | Name | Decision | Date |
|---|---|---|---|
| Product owner | To be assigned | Pending review | |
| Engineering owner | To be assigned | Pending review | |
| Security / privacy reviewer | To be assigned | Pending review | |
| QA owner | To be assigned | Pending review | |

## How to Read This Specification

This document specifies the product to be built, not a claim that the product already exists. Requirements are written to be testable. Items labelled “proposed default” are the starting policy for the pilot and must be configurable until evidence supports a final value. Implementation technology is intentionally not locked unless a requirement depends on it.

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### Specification boundary

Routine employee development operations are intended to run autonomously after initial policy and identity configuration. The platform is for learning, skill development, and internal gamification; it is not a clinical IQ instrument and must not be the sole basis for high-impact employment decisions.

# 1. Purpose and Intended Use

This MSRS defines the requirements for DREAM TEAM AI Development OS, an AI-managed employee development platform with personalized training, assessment, practical challenges, gamified progression, competition, durable records, and administrator analytics. It is the single master pre-development reference for product behaviour and acceptance.

## 1.1 Purpose

- Turn employee development into a continuous, measurable loop rather than a one-time test or a static course catalogue.
- Develop all ten agreed dimensions while tailoring practice to Sales, Technical, and Admin work.
- Use AI to autonomously diagnose needs, create and validate new challenges, teach mistakes, adapt difficulty, maintain records, run competitions, and issue qualifying level-ups.
- Preserve the complete history of assessments, answers, failures, feedback, task states, achievements, and promotion evidence.
- Give authorized administrators a complete, evidence-linked view of employee and team development without requiring them to grade every task or manually operate the training programme.

## 1.2 Intended Use

The system is intended for internal training and development of DREAM TEAM SERVICES personnel. It supports onboarding, baseline assessment, daily practice, role-based scenarios, simulated tasks, workplace improvement missions, team competitions, retention checks, and longitudinal reporting.

## 1.3 Measurement Disclaimer

**Important distinction**

The application may use “Intelligence” in its product name, but its scores are estimates of demonstrated performance on defined tasks. They are not a validated IQ score, a medical/psychological diagnosis, or a complete measure of a person’s worth or potential.

# 2. Product Vision, Goals and Boundaries

## 2.1 Product Vision

Create a continuously operating AI development environment in which every employee receives a personalized path to become more capable, independent, adaptable, and effective at work. Progress is earned through demonstrated mastery and practical outcomes, not tenure, clicks, or accumulated easy points.

## 2.2 Success Outcomes

- Every active employee has a complete profile and a current evidence-based view across all ten dimensions.
- Every employee is assigned to an appropriate role track: Sales, Technical, or Admin; tracks have distinct assessments and task rubrics.
- Every newly delivered assessment item passes answer, clarity, difficulty, safety, and novelty checks.
- No employee is served an exact or substantially equivalent repeat question from their own exposure history.
- Level progression is automatically calculated from configured mastery gates and retained evidence.
- All attempts are saved centrally and can be inspected chronologically by the employee and authorized administrators.
- AI can operate routine training and competition flows without day-to-day human grading, content authoring, or scheduling.
- Progress is assessed using fresh questions and real or verifiably simulated work, not only a single test score.

### 2.3 In Scope

- Employee identity, onboarding, track assignment, profile setup, and baseline diagnosis.
- Ten-dimension assessments, AI-generated item variants, duplicate prevention, objective scoring, rubric scoring, feedback, remediation, and retesting.
- Personalized daily quests, weekly role challenges, team missions, delayed retention checks, and monthly/quarterly reassessment plans.
- Six ranks, XP, skill mastery, achievements, seasons, role-aware leaderboards, team competitions, and auditable level-ups.
- Employee assessment archive and personal analysis; admin analytics across individuals, tracks, dimensions, attempts, question quality, and progress.
- Scheduled jobs, event processing, notifications, retries, audit logging, exports, backups, and system health monitoring.

### 2.4 Out of Scope Unless Separately Approved

- Clinical IQ tests or claims of psychological diagnosis.
- Automatic salary changes, hiring/firing decisions, disciplinary decisions, or employment termination based only on AI scores.
- Biometric emotion detection, facial-expression profiling, or personality inference from voice/video.
- Assuming every Technical employee must code. Coding challenges are included only when a specific job requires coding.
- Replacing the company's CRM, project management, HRIS, or source-control platform. Integrations may be added where approved.
- Guaranteeing a particular cognitive improvement rate or rank distribution before the system has pilot evidence.

## 3. Terms and Definitions

| Term | Definition |
|---|---|
| **Dimension** | One of the ten official capability areas specified in Section 6. |
| **Track** | Role-specific learning and assessment context: Sales, Technical, or Admin. |
| **Challenge / task** | A question, scenario, practical activity, simulation, or deliverable used for development. |
| **Assessment attempt** | An immutable record of one employee's response to an assessment or task. |
| **Question identity** | A stable identifier/version plus normalized content and similarity fingerprints used to prevent repeats. |
| **XP** | Experience points used for engagement and progression display; XP alone does not establish mastery. |
| **Mastery** | Evidence-based estimate that an employee can perform a specified skill at the required quality level. |
| **Competitive rating** | Season-scoped measure used to compare performance in a defined league. |
| **Rank / level** | One of the six progression tiers, unlocked by meeting configured gates. |
| **Retention check** | A later assessment using fresh content to verify that learning persisted. |
| **AI autonomy** | Routine operation by AI services and deterministic software rules without manual grading or manual task assignment. |
| **Human involvement** | Initial system ownership/configuration or optional audit/appeal. It is not a prerequisite for normal daily learning operations. |
| **Near-duplicate** | A question that is substantially the same in wording, scenario, figures, answer logic, or required solution despite superficial changes. |

| Term | Definition |
|---|---|
| **Verified outcome** | An outcome supported by a deterministic check, trusted integrated event, structured evidence, or validated artifact—not merely an AI assertion. |
| **Admin track** | Learning path for employees doing administrative/operational work. Distinct from platform administrator permissions. |

# 4. Users, Roles and Access Model

| Role | Primary capabilities | Restrictions |
|---|---|---|
| **Employee / learner** | Complete assessments, practise, view own profile/history, receive coaching, submit task evidence, join eligible competitions. | Cannot alter scores, XP ledger, question history, rank gates, other employees' records, or system rules. |
| **Sales-track learner** | Employee capabilities plus Sales-specific missions and analytics. | No automatic access to unrelated employees' full records. |
| **Technical-track learner** | Employee capabilities plus non-coding technology tasks by default; job-specific coding only when assigned. | Cannot access privileged infrastructure or actual production systems through a training challenge unless explicitly sandboxed and authorized. |
| **Admin-track learner** | Employee capabilities plus operations, planning, reporting and process-improvement tasks. | Does not gain platform administrator access solely by being assigned the Admin learning track. |
| **Platform administrator** | Configure organization, roles, tracks, policies, thresholds, integrations, access, and review the complete authorized employee analytics. | Configuration and administrative actions are logged. Cannot silently rewrite historical attempts. |
| **System service / AI agent** | Generate and validate content, assign tasks, grade, coach, calculate XP/mastery/ranks, schedule jobs, write reports, and monitor failures. | Must act through scoped service identities and enforce all deterministic policy checks. |

## 4.1 Permission Principles

- Access is denied by default and granted by explicit role and organization scope.
- Employees can access their own detailed history. Administrators can access the records within their authorized scope.
- All administrative changes to roles, scoring settings, thresholds, questions, or historical corrections create audit events.
- AI agents use least-privilege service identities; prompt text cannot grant permissions or override authorization rules.
- Exports and downloads require authorization and are logged when they contain personal employee data.

# 5. Product Overview and Operating Principles

## 5.1 Product Modules

| Module | Responsibility |
|---|---|
| **AI Intelligence Director** | Coordinates diagnosis, next-task selection, difficulty adaptation, remediation, retention scheduling, and reports. |

| Module | Responsibility |
|---|---|
| **Assessment Engine** | Creates assessment sessions, serves validated questions, collects answers, and records attempts. |
| **Question Intelligence Service** | Generates, normalizes, deduplicates, validates, versions, and retires questions. |
| **AI Coach** | Explains mistakes, provides progressive hints, teaches methods, and generates fresh transfer exercises. |
| **Mastery Engine** | Maintains dimension-level evidence and determines eligibility for level promotion. |
| **Gamification Engine** | Maintains XP ledger, achievements, streaks, seasons, leaderboards, and rank-up experiences. |
| **Challenge Engine** | Manages role-specific practical tasks, deadlines, submission evidence, verification, and completion states. |
| **Employee Profile & History** | Stores the complete chronological record and exposes understandable progress analysis. |
| **Admin Intelligence Center** | Aggregates individual, track, team, content-quality, and learning-effectiveness analytics. |
| **Automation & Reliability** | Runs scheduled work, retries, idempotent events, health monitoring, backup, and incident handling. |

### 5.2 Foundational Product Rules

- Practice may earn XP; mastery earns ranks; league performance determines seasonal standings. Keep these values distinct.
- Failed attempts are learning evidence and remain in history. They do not erase previous achievements or earned XP.
- The exact same question must never be served twice to the same employee. Substantially equivalent variants are also prohibited for that employee.
- The underlying learning objective can recur only in a meaningfully different scenario, question form, or application context.
- Routine operations must not depend on an administrator opening the app or manually assigning/grading daily tasks.
- Every important AI conclusion shown to an employee or administrator must link to the attempt, evidence, rubric, or rule that supports it.
- The system must distinguish “not passed”, “not completed”, “expired”, and “invalid due to platform error”.
- The platform is designed for employee development. Competition must not reward shortcuts that undermine learning, safety, fairness, or record integrity.

## 6. Official Ten-Dimension Framework

The product uses exactly the ten dimensions below. No additional top-level intelligence dimensions are introduced. Skills and subskills may exist within a dimension for measurement and training, but the official framework remains these ten.

| No. | Dimension | What it measures | Example assessment |
|---|---|---|---|
| **1** | Cognitive Ability | How quickly and accurately a person understands and processes information. | Explain a new concept and apply it to a novel situation. |
| **2** | Logical Reasoning | Ability to identify patterns and draw logical conclusions. | Solve a logic puzzle or infer what follows from a set of facts. |
| **3** | Learning Agility | How quickly a person learns unfamiliar skills and transfers learning. | Teach an unfamiliar tool or method and assess use on a new task. |

| No. | Dimension | What it measures | Example assessment |
|---|---|---|---|
| 4 | Critical Thinking | Ability to evaluate information, assumptions, claims, and evidence. | Critique a campaign, vendor claim, report, or proposed solution. |
| 5 | Problem-Solving | Ability to identify root causes and develop effective solutions. | Investigate a realistic business or workflow failure and propose a fix. |
| 6 | Creativity & Innovation | Ability to generate original, useful, relevant alternatives. | Develop multiple distinct concepts or improve a process under constraints. |
| 7 | Decision-Making | Ability to choose effectively under constraints, uncertainty, and trade-offs. | Choose a response when time, cost, quality, risk, and resources conflict. |
| 8 | Communication | Ability to explain ideas clearly and understand what others need. | Explain a solution to a non-specialist and respond to clarifying questions. |
| 9 | Emotional Intelligence | Ability to interpret social context, manage reactions, and work constructively with people. | Respond to a frustrated customer or disagreement using a structured scenario. |
| 10 | Execution & Adaptability | Ability to turn plans into results and adjust when requirements change. | Complete a task to a standard and adapt to a change in requirements. |

### 6.1 Measurement Policy

- Each dimension has its own score/evidence history; the role-weighted total must not hide a critical weak dimension.
- Assessments should combine objective tests, structured scenarios, role-related tasks, and evidence across time.
- Self-ratings may be collected for reflection but must remain separate from assessed performance.
- Emotional Intelligence is measured with structured scenarios and demonstrated behaviours, not medical claims or biometric emotion recognition.
- Assessment language, difficulty, time limit, hints, and permitted resources must be recorded so scores can be interpreted fairly.
- Performance estimates must include sample size and confidence/quality status where evidence is insufficient.

## 7. Role-Specific Development Tracks

All tracks develop the same ten dimensions but use different scenarios, examples, rubrics, and mastery evidence. Track configuration is selected from the employee's actual work, not inferred solely from a title.

### 7.1 Sales Development Track

Purpose: improve the ability to understand customers, qualify leads, communicate value, handle objections, prioritize follow-ups, make trade-offs, and turn sound decisions into measurable sales execution.

- Cognitive Ability: extract key customer needs from a brief or conversation summary.
- Logical Reasoning: analyse funnel data and identify which conclusions the numbers support.
- Learning Agility: learn a new qualification or follow-up framework and apply it to an unseen prospect.
- Critical Thinking: challenge assumptions about why a prospect did not convert.
- Problem-Solving: diagnose lead leakage using evidence such as response time, qualification, and objections.
- Creativity & Innovation: propose distinct value propositions or outreach approaches without inventing offers.
- Decision-Making: prioritize prospects under limited time and explain trade-offs.
- Communication: explain value clearly and respond appropriately to simulated objections.
- Emotional Intelligence: respond constructively to frustration, hesitation, or disagreement in a scenario.
- Execution & Adaptability: complete a follow-up plan, respect deadlines, and adapt to changed lead information.

### 7.2 Technical Development Track (Non-Coding by Default)

**Explicit requirement**

"Technical" means technology-related work and does not mean coding by default. The track focuses on understanding and using technology, investigating issues, making technical decisions, and improving workflows. Coding is included only where the specific employee role requires it.

- Cognitive Ability: learn a new software tool from instructions and complete a novel task.
- Logical Reasoning: diagnose a process failure from the order of steps and the available evidence.
- Learning Agility: learn an unfamiliar AI, CRM, networking, project-management, or productivity tool.
- Critical Thinking: compare software products and identify unsupported vendor claims.
- Problem-Solving: investigate a slow website, connectivity issue, access problem, or broken workflow using evidence.
- Creativity & Innovation: design a more efficient workflow with existing software, AI tools, or no-code automation.
- Decision-Making: compare options across cost, usability, security, reliability, and delivery time.
- Communication: explain a technical issue and resolution to a non-technical colleague or customer.
- Emotional Intelligence: respond calmly and constructively when a customer or teammate is frustrated by a technical issue.
- Execution & Adaptability: configure a tool in an approved environment, test the result, document steps, and adapt to changed requirements.

### 7.3 Admin / Operations Development Track

Purpose: improve planning, interpretation of information, process management, coordination, prioritization, reporting, risk awareness, and execution across business operations.

- Cognitive Ability: interpret an unfamiliar report and summarize the important information.
- Logical Reasoning: detect inconsistencies in schedules, resource plans, or operational data.
- Learning Agility: learn a new business workflow or management application and use it correctly.
- Critical Thinking: separate facts from assumptions in a proposed policy or management update.
- Problem-Solving: identify root causes of repeated delays, errors, or hand-off failures.
- Creativity & Innovation: redesign a process to reduce effort or prevent mistakes.
- Decision-Making: allocate limited resources and document trade-offs and risks.
- Communication: communicate instructions, decisions, and status unambiguously.
- Emotional Intelligence: resolve workplace misunderstandings and respond constructively to difficult conversations.
- Execution & Adaptability: complete an operational task on time and update the plan when a requirement changes.

### 7.4 Initial Track Assignment

At onboarding, the AI should ask structured questions about the employee's responsibilities and use approved role/team records to suggest a track. Track assignment must be stored with effective dates. If the person performs mixed duties, the profile may attach more than one task domain while maintaining a primary track for comparable reporting.

## 8. AI Autonomy and Decision Model

### 8.1 Autonomy Objective

**Operating objective**

After initial organization, identity, role, and policy configuration, the platform should run routine assessment, training, grading, coaching, task assignment, progression, competitions, reporting, and history maintenance with zero required day-to-day human involvement.

Autonomy does not mean unbounded model authority. The AI plans and creates learning activities; deterministic services enforce identity, due dates, scoring formulas, question-exposure rules, XP transactions, promotion gates, access controls, and audit integrity. Owners may inspect or override the system, but routine training must not depend on such involvement.

### 8.2 AI Decision Responsibilities

| AI responsibility | Inputs | Required output / guardrail |
|---|---|---|
| **Diagnose** | Validated attempt history, skill estimates, track, employee goals, completion status. | Ranked skill gaps with evidence links; avoid unsupported personal labels. |
| **Choose next activity** | Skill gap, recent practice, due items, difficulty, fatigue/length policy. | One suitable task, learning objective, difficulty, target time, and rationale. |
| **Generate questions** | Task blueprint, track, dimension, difficulty, prior exposure, language. | Candidate questions plus correct answers, rubric, explanation, and source/logic metadata. |
| **Validate content** | Candidate item, answer key, item policy, duplicate search results. | Pass/reject decision; reject ambiguous, unsolved, unsafe, inaccurate, or repeated items. |
| **Score** | Response, item answer key/rubric, task evidence, configured rules. | Score, criterion-level evidence, feedback, confidence/validation state, and reason codes. |
| **Coach** | Failed criteria, answer, rubric, known skill gaps. | Specific explanation, progressive hints, next learning step; do not expose answer before attempt policy permits. |
| **Promote** | Mastery records, fresh assessments, task evidence, retention checks, rank policy. | Eligibility result and rule-by-rule evidence; update rank only if all gates pass. |
| **Monitor system** | Job queues, error logs, latency, AI validation failures, duplicate rejection rate. | Retry, use fallback, pause affected workflow, or record a recoverable incident. |

### 8.3 Uncertain AI Results

- If a generated answer key, rubric, or question cannot be validated, the system must reject it and generate a replacement.
- For subjective work, use structured rubrics and an independent second AI judge for configured high-impact or low-confidence assessments.
- If automated graders disagree beyond the configured tolerance, trigger an additional AI adjudication or mark the result as “validation pending”; do not fabricate a confident score.
- Platform-generated invalid attempts must not count as employee failures. Record the system fault and provide a fresh assessment.
- Administrative overrides are optional controls, not a dependency for normal operation. Every override must be attributable and logged.

## 9. Primary User and System Workflows

### 9.1 Employee Onboarding and Baseline

1. Create or invite the employee identity and attach the organization/team and primary track.
2. Explain the ten dimensions, privacy, assessment rules, time expectations, score meaning, and competition policy.
3. Run a baseline using fresh questions across all ten dimensions, with tasks appropriate to the track.
4. Calculate dimension-level results, mark sample sizes and uncertainty, and generate an individual starting profile.
5. Create a personalized development plan and assign the first tasks. Do not force an employee to grind through low-level questions if the baseline proves existing competence.

### 9.2 Daily Training Loop

1. The scheduler selects a due session based on the employee plan, available time, current skill gaps, and task history.
2. The question service generates or selects an unseen, validated item matched to the target learning objective.
3. The employee completes the task; the platform records response, timings, permitted hints, and submission state.
4. The scoring engine evaluates the result and stores the complete attempt before displaying the result.
5. The AI coach provides tailored feedback and schedules a different follow-up exercise or retention check.
6. XP, mastery, badges, and rank eligibility are updated through separate rule-controlled transactions.
7. The profile and admin analytics update asynchronously and idempotently.

### 9.3 Failure and Recovery

1. Classify the result as failed, incomplete, expired, invalid/platform error, or awaiting automated validation.
2. Persist the attempt and reason; do not overwrite or delete a previous attempt.
3. Diagnose the specific learning gap and teach the relevant concept or method.
4. Create a different question and scenario targeting the same underlying skill.
5. If validation or generation fails, retry within configured limits and use a validated unseen fallback bank.
6. Schedule a later retention check using a new item. Grant mastery only when the required criteria pass.

### 9.4 Rank-Up

When an employee meets a level gate, the Mastery Engine evaluates the evidence using a deterministic rule set. It records a promotion event containing the prior level, new level, policy version, qualifying assessment IDs, challenge outcomes, retention evidence, and timestamp. The UI celebrates the promotion and unlocks the next tier. A single XP threshold cannot bypass mastery gates.

### 9.5 Competition Round

The Competition Engine schedules a round, defines eligible participants, issues unique but equivalently difficult variants, enforces the time and resource policy, scores with the same rubric version for the round, stores attempts, recalculates the standings, and publishes results. The platform records season snapshots so later score recalculations do not silently rewrite past standings.

## 10. Functional Requirements

MUST requirements are release-blocking. SHOULD requirements are expected unless a documented product decision defers them. Every requirement is assigned a stable ID for implementation and testing.

### 10.1 Identity, Profile and Track

**FR-001 Authenticated employee identity** [MUST]

Each employee MUST have a stable unique user ID, display identity, organization/team association, active state, primary track, and creation timestamp. Shared anonymous profiles are not acceptable for company-wide records.

**Acceptance:** *Two employees cannot overwrite or view each other's data by changing a client-side identifier.*

**FR-002 Role separation** [MUST]

The system MUST distinguish employee access, Sales/Technical/Admin learner tracks, platform administrator permissions, and service-agent permissions. An Admin learning track MUST NOT automatically grant administrator privileges.

**Acceptance:** *A track assignment changes training content but does not grant additional access.*

**FR-003 Employee profile** [MUST]

Each employee MUST have a profile showing current rank, XP, mastery by dimension, primary track, assigned goals, due tasks, achievements, competition history, and a chronological assessment archive.

**Acceptance:** *Profile values reconcile with saved attempt and ledger records.*

**FR-004 Initial diagnosis [MUST]**

The system MUST create a baseline across all ten dimensions using the appropriate role track and mark insufficient evidence rather than invent a score.

**Acceptance:** *A new profile identifies evidence collected and dimensions not yet reliably measured.*

**FR-005 Track changes [SHOULD]**

Authorized track changes MUST be effective-dated; historical attempts retain the track and rubric under which they were assessed.

**Acceptance:** *A track change does not retroactively alter old scores.*

## 10.2 Question Generation, Validation and Uniqueness

**FR-006 AI-generated challenge content [MUST]**

The Question Intelligence Service MUST generate varied question types matched to dimension, track, level, difficulty, language, and learning objective. Every item must include an expected answer, scoring rubric, explanation, difficulty metadata, and validation state.

**Acceptance:** *No candidate becomes live until it passes configured validation.*

**FR-007 Answer-key validation [MUST]**

Objective items MUST have a deterministic answer key. Open-ended items MUST have explicit rubric criteria, scoring anchors, and examples or constraints sufficient for consistent automated evaluation.

**Acceptance:** *Invalid or ambiguous answer keys are rejected before delivery.*

**FR-008 Per-employee non-repetition [MUST]**

The system MUST prevent an employee from receiving any exact question they have previously seen, including across baseline, daily practice, retests, rank assessments, and competitions.

**Acceptance:** *Exposure history is checked before issuing every item.*

**FR-009 Near-duplicate rejection [MUST]**

The system MUST normalize text and compare semantic content, scenarios, numbers, choices, answer logic, and solution paths to reject substantially equivalent questions for that employee. Superficial word changes do not make a question new.

**Acceptance:** *A known paraphrase or same-solution variant is rejected in automated tests.*

**FR-010 Fresh competition variants [MUST]**

Participants in a competitive round MUST receive distinct assessment items of equivalent difficulty and the same rubric version. Shared templates may be used only to generate distinct instances.

**Acceptance:** *No participant can see another participant's variant or answer before the round closes.*

**FR-011 Fallback item supply [MUST]**

If AI cannot produce a valid new item within the retry budget, the system MUST use a validated unseen fallback item or defer the session with a transparent system status. It MUST NOT recycle a known item.

**Acceptance:** *Generation failure never leads to a repeated question or fabricated pass/fail result.*

**FR-012 Question provenance and versioning [MUST]**

Each item MUST record a stable ID, version, prompt/content, answer key or rubric, track, dimension, difficulty, model/prompt version where applicable, validation results, similarity fingerprint, and exposure records.

**Acceptance:** *The historical item can be reconstructed for audit.*

**FR-013 Question quality calibration [SHOULD]**

The system SHOULD estimate item difficulty, ambiguity, discrimination and observed completion time from validated attempts, then flag poor items for automatic quarantine.

**Acceptance:** *Invalid or misleading questions are excluded from mastery calculations pending revalidation.*

## 10.3 Assessment and Scoring

**FR-014 Multi-format assessments** [MUST]

Assessments MUST support objective questions, reasoning explanations, scenario decisions, multi-step problem-solving, learning-transfer tasks, creative responses, communication exercises, and role-specific practical deliverables.

**Acceptance:** *All ten dimensions are covered by defined item blueprints.*

**FR-015 Consistent attempt recording** [MUST]

Each attempt MUST store the question version, response, answer/criterion scores, total, start/end timestamps, allowed hints, task state, rubric version, feedback, and validation status.

**Acceptance:** *An attempt is stored before results are shown to the learner.*

**FR-016 Deterministic scoring where possible** [MUST]

Objective answers, arithmetic, required-step checks, task deadlines, XP rules, rank gates, and promotion eligibility MUST be calculated by deterministic code rather than free-form AI text.

**Acceptance:** *Changing the model response cannot change deterministic results.*

**FR-017 Rubric-based scoring** [MUST]

Open-ended work MUST be evaluated against versioned criteria with criterion-level notes and supporting evidence. Scores cannot rely solely on length, confidence, style, or unsupported AI impressions.

**Acceptance:** *A stored score explains which criteria passed or failed.*

**FR-018 Automated second evaluation** [MUST]

For subjective assessments configured as high-impact or low-confidence, a separate evaluation pass MUST check the first score. Disagreements beyond a defined tolerance trigger automated adjudication or validation-pending status.

**Acceptance:** *The system does not silently force a score when judges materially disagree.*

**FR-019 Confidence and evidence quality** [MUST]

Each automated evaluation MUST record a confidence/quality state and evidence basis. Low evidence must be represented as low confidence or insufficient data, not as a fabricated precise measurement.

**Acceptance:** *Insufficient evidence does not unlock a mastery gate.*

**FR-020 Fresh assessment forms** [MUST]

Baseline and later assessments MUST use distinct unseen questions and equivalent blueprints where scores are compared over time.

**Acceptance:** *Improvement reporting does not depend on repeating the same question.*

## 10.4 AI Coach and Adaptive Training

**FR-021 Personalized development plan** [MUST]

The AI Director MUST select development priorities using the employee's history, skill estimates, failed criteria, role track, completed tasks, retention and configured goals.

**Acceptance:** *Each assigned activity links to a target dimension and reason.*

**FR-022 Mistake diagnosis** [MUST]

After a failed or weak response, the AI MUST identify the specific reasoning or execution gap and give actionable feedback. It SHOULD distinguish a knowledge gap, assumption error, evidence error, calculation error, decision trade-off, communication issue, or incomplete execution.

**Acceptance:** *Feedback references the response and rubric rather than generic praise.*

**FR-023 Progressive hints** [SHOULD]

The Coach SHOULD provide hints in stages, preserving the opportunity to solve the task before revealing a complete explanation according to the configured assessment policy.

**Acceptance:** *Hint use is recorded and considered by the applicable scoring rules.*

**FR-024 Adaptive difficulty [MUST]**

Difficulty MUST adjust based on validated performance, including increasing complexity after demonstrated mastery and offering scaffolding after repeated difficulty.

**Acceptance:** *Adaptation never changes the official scoring rubric midway through an attempt.*

**FR-025 Learning transfer [MUST]**

The Coach MUST periodically assess the same learning objective in a different context to verify transfer rather than memorization.

**Acceptance:** *A transfer check uses a distinct, validated scenario.*

**FR-026 Spaced retention checks [MUST]**

The scheduler MUST schedule later fresh assessments for priority skills to verify retention. Timing is configurable and recorded.

**Acceptance:** *A temporary immediate success is not the only evidence used for higher-rank mastery.*

**FR-027 Bounded AI loops [MUST]**

Generation, grading, repair and adjudication loops MUST have configured retry, time, and cost limits. Exhaustion results in a clear recoverable state and fallback behaviour.

**Acceptance:** *No unbounded retry or API-credit loop is possible.*

## 10.5 Tasks, Deadlines and Execution

**FR-028 Task lifecycle [MUST]**

Tasks MUST support at least Assigned, In Progress, Completed, Passed, Failed, Expired, Cancelled, and Invalid/System Error states with defined transitions and timestamps.

**Acceptance:** *A task state cannot be changed silently or without a reason event.*

**FR-029 Deadline enforcement [MUST]**

The backend MUST enforce due dates and configured grace policies. Failure to complete a valid required task by its deadline records the relevant failure state and withholds its mastery credit.

**Acceptance:** *Client-side clock manipulation cannot change a deadline result.*

**FR-030 Platform-error protection [MUST]**

Network outages, service faults, invalid generated content, or broken assessment sessions MUST be distinguishable from employee failure and MUST NOT unjustly reduce mastery.

**Acceptance:** *System-error attempts have explicit reason codes and do not count as learner failures.*

**FR-031 Evidence-based completion [MUST]**

Practical tasks MUST define evidence of completion, such as a passed automated check, validated artifact, required workflow event, structured submission, or test result. AI-only assertion is insufficient for outcomes that can be verified mechanically.

**Acceptance:** *Completion status links to the evidence record or validation rule.*

**FR-032 Task adaptation [SHOULD]**

When circumstances or requirements change, the system SHOULD issue an updated variant and record the changed requirement rather than silently changing an active task.

**Acceptance:** *The learner can see which requirement changed and when.*

## 10.6 XP, Mastery, Ranks and Achievements

**FR-033 Separate progression ledgers [MUST]**

The system MUST maintain separate XP, mastery evidence, current rank, and competitive rating. XP and participation cannot by themselves unlock a rank.

**Acceptance:** *A user with high XP but failing mastery gates is not promoted.*

**FR-034 XP ledger [MUST]**

XP changes MUST be stored as immutable transactions with reason, source attempt/task, rule version, timestamp, and idempotency key. Duplicate events cannot award XP twice.

**Acceptance:** *Replaying an event does not duplicate the transaction.*

**FR-035 Six official ranks [MUST]**

The rank ladder MUST be Basic, Intermediate, Advanced, Pro, Pro Max, and Ultra Pro Max in that order. Labels and rules are centrally configured.

**Acceptance:** *The UI, profile, admin dashboard, APIs and reports agree on rank state.*

**FR-036 Rank-up eligibility [MUST]**

Promotion MUST require all configured gates: fresh assessments, critical-dimension minimums, practical challenge evidence, learning transfer and, for higher tiers, retention/capstone evidence.

**Acceptance:** *No single score or XP total bypasses any required gate.*

**FR-037 Automatic promotion event [MUST]**

When all gates pass, the system MUST promote automatically and record policy version, old/new rank, qualifying records, and timestamp.

**Acceptance:** *A promotion is reproducible from stored evidence and rules.*

**FR-038 Permanent achievement history [MUST]**

Earned badges, rank-ups, personal bests and season awards MUST persist independently of seasonal score resets.

**Acceptance:** *Season reset does not delete career history.*

**FR-039 Failure handling [MUST]**

A failed task withholds the associated mastery credit and schedules targeted remediation. It MUST NOT erase prior earned XP or assessment history.

**Acceptance:** *The failed attempt remains visible after a later successful retest.*

**FR-040 No artificial rank decay [SHOULD]**

Rank should not drop solely because time passed or a new attempt failed. A separately defined “current mastery confidence” may decrease when evidence becomes stale, without deleting prior achievements.

**Acceptance:** *Historical rank events remain immutable.*

## 10.7 Competition and Social Learning

**FR-041 Daily quests [MUST]**

The system MUST issue short role-appropriate practice missions, with the target dimension, estimated time, difficulty, due state and reward disclosed.

**Acceptance:** *Eligible learners receive scheduled tasks without manual assignment.*

**FR-042 Weekly leagues [SHOULD]**

The Competition Engine SHOULD run configured leagues with eligible entrants, distinct variants, equivalent difficulty, deadline rules and auditable scores.

**Acceptance:** *Standings can be reconstructed from attempt records and the league rule version.*

**FR-043 Multiple leaderboards [MUST]**

At minimum, support Mastery, Most Improved, Execution, and Team Contribution leaderboards. Keep the underlying metric and eligibility rules visible.

**Acceptance:** *A participant can see how each board is calculated.*

**FR-044 Season management [SHOULD]**

Seasons SHOULD have defined start/end dates and independent standings. Seasonal resets MUST NOT reset mastery, assessment history, rank events or permanent achievements.

**Acceptance:** *A past season snapshot remains available.*

**FR-045 Team missions [MUST]**

Monthly or configured team missions MUST define a shared objective, distinct participant contributions, evidence, scoring rubric, and validated outcome measures.

**Acceptance:** *Team reward is tied to evidence of contribution and outcome, not only submission count.*

**FR-046 Anti-farming [MUST]**

The system MUST limit repeated easy-task XP farming, duplicated submissions, multi-account abuse, and unsupported completion claims.

**Acceptance:** *XP and ratings are produced by server-validated, idempotent rules.*

**FR-047 Safe competition [MUST]**

Competition MUST avoid public humiliation, exposure of private detailed answers, or punitive public failure rankings. Employees may see their own details; published standings show only approved metrics and identity fields.

**Acceptance:** *The privacy/access rules apply to every leaderboard and announcement.*

## 10.8 Employee History and Admin Analytics

**FR-048 Complete assessment archive [MUST]**

For each employee, permanently retain every assessment/test/task attempt, question version, submitted answer, score, criterion-level result, feedback, failure reason, completion timestamps, and associated retests, subject to the approved retention policy.

**Acceptance:** *A new attempt appends a record and never overwrites its predecessor.*

**FR-049 Employee progress analysis [MUST]**

The profile MUST show baseline-to-current trends, dimension-level strengths/gaps, recent results, retention, rank history, task completion, feedback, badges and the next development plan.

**Acceptance:** *Each chart/summary can drill through to source attempts.*

**FR-050 Admin company overview [MUST]**

Authorized administrators MUST be able to review counts by track/rank, participation, completion and failure states, skill gaps, improvement trends, training effectiveness, question quality and practical outcomes.

**Acceptance:** *Filters and aggregates match the underlying authorized records.*

**FR-051 Individual admin drill-down [MUST]**

Administrators MUST be able to inspect an individual's full authorized assessment and development history, including failed attempts and promotion evidence.

**Acceptance:** *Admin drill-down preserves chronology and supports filtering/export.*

**FR-052 Track-aware comparison [MUST]**

Sales, Technical and Admin performance MUST be analyzed against track-appropriate tasks and weight profiles. Cross-track comparisons may display common dimensions but MUST disclose differences in rubrics and tasks.

**Acceptance:** *A sales simulation score is not silently treated as equivalent to a technical challenge score.*

**FR-053 AI management reports [MUST]**

The system MUST generate scheduled summaries of progress, overdue tasks, repeated skill gaps, training effects, system/content quality issues and next actions. Every material statement must reference source records or be marked as a hypothesis.

**Acceptance:** *An admin report can trace conclusions to records and dates.*

**FR-054 Export and backup [MUST]**

Authorized users MUST be able to export appropriate employee records and reports in supported formats. Backups must be restorable and checked. Export permissions and operations are logged.

**Acceptance:** *A restore test confirms identity, history and ledger consistency.*

## 10.9 Administration, Notifications and Reliability

**FR-055 Policy configuration [MUST]**

Platform administrators MUST be able to configure track mappings, approved weight profiles, rank gates, XP rules, season dates, task cadence, retry budgets, duplicate thresholds and privacy settings through a controlled configuration layer.

**Acceptance:** *A change takes effect according to a versioned effective time; old attempts retain old policy versions.*

**FR-056 Notifications** [SHOULD]

The system SHOULD notify employees of new tasks, deadlines, feedback, rank-ups and competitions using approved channels. Notification frequency and quiet hours must be configurable.

**Acceptance:** *Repeated events are deduplicated and notifications can be traced to source events.*

**FR-057 Scheduled autonomous operation** [MUST]

Backend schedulers MUST run planned tasks, retention checks, season transitions, reports and routine workflows even when the employee or administrator is offline.

**Acceptance:** *A closed browser does not prevent scheduled backend tasks from running.*

**FR-058 Health monitoring** [MUST]

The platform MUST monitor AI provider availability, queue age, failed jobs, repeated invalid generation, scoring-adjudication disagreement, duplicate rejections, storage errors and latency.

**Acceptance:** *Critical system faults are recorded and trigger automated recovery or safe pause behaviour.*

**FR-059 Idempotent processing** [MUST]

Retries and repeated events MUST NOT duplicate attempts, XP, promotions, badges, notifications or competition results.

**Acceptance:** *Reprocessing the same event is safe and produces no duplicate side effect.*

**FR-060 Audit trail** [MUST]

The system MUST record access-relevant actions, policy changes, role changes, admin exports, score corrections, adjudications and promotion decisions with actor/service identity and timestamp.

**Acceptance:** *Audit records cannot be silently edited by ordinary application users.*

# 11. Business Rules and Scoring Model

## 11.1 Ten-Dimension Scores

Each dimension is reported on a 0–100 scale derived from validated assessment evidence. Each task uses a rubric version that maps observed criteria to the dimension. Confidence and sample size must be displayed where available. Scores are updated only through stored evidence; an AI summary cannot directly overwrite a score.

## 11.2 Role-Weighted Summary

A role-weighted 0–100 summary may be calculated as the sum of each dimension score multiplied by its role weight. Proposed defaults are in Appendix A. Weights are product settings, must total 100%, and require pilot calibration. The summary is for development and analytics; it must not hide minimum gates for critical dimensions.

## 11.3 Illustrative Mastery Calculation

A candidate mastery estimate can combine recent validated performance, difficulty, evidence quality, task completion, transfer success and retention. The exact estimator must be documented, deterministic where possible, versioned, and tested. It must not treat question count alone as mastery. The first release may use transparent weighted evidence rules rather than an opaque model, provided requirements are met.

## 11.4 Proposed Rank Gates

Appendix B gives proposed baseline thresholds. In addition to score thresholds, promotions require fresh distinct assessment evidence and practical/retention requirements appropriate to the rank. A failed or missing critical criterion blocks promotion even when the overall score exceeds the threshold.

### 11.5 Completion and XP

| Event | XP policy | Mastery policy |
|---|---|---|
| **Valid practice completed** | May award modest XP for verified effort and completion. | Mastery change depends on answer quality, not participation alone. |
| **Correct objective response** | Award according to difficulty and rule version. | Update the relevant skill evidence. |
| **Successful practical task** | Award a higher configured reward when independently verified. | Counts toward practical mastery only if rubric gates pass. |
| **Failed task** | May award limited participation XP only if policy explicitly allows it; no pass reward. | No mastery credit for criteria not met; retain attempt and trigger remediation. |
| **Invalid/system-error attempt** | No punitive XP penalty. | Excluded from learner failure rates and rank gates; record system fault. |
| **Mentorship/team contribution** | Reward only when contribution evidence is validated. | Can contribute to collaboration-related evidence when supported by an approved rubric. |

### 11.6 Fairness Rules

- Comparable assessments must use equivalent difficulty blueprints, rubric versions, time limits, and permitted-resource rules.
- Language or accent must not be penalized when it is irrelevant to the skill being assessed.
- Accessibility accommodations and permitted time adjustments must be recorded, without exposing unnecessary personal details.
- Training volume, attendance, speed, and XP alone are not proof of competence.
- The system must show a clear reason when a task fails or promotion is denied, and allow automated re-evaluation when scoring checks fail.
- Historic records remain available even when a policy, rubric, model, or rank threshold changes.

## 12. Data Requirements and Record Model

The following is the minimum logical data model. Physical database selection and deployment topology are implementation decisions, but central persistent storage is required for shared histories and autonomous processing.

| Entity | Required fields / purpose |
|---|---|
| **EmployeeProfile** | userId, displayName, teamId, primaryTrack, role metadata, status, locale, accessibility settings, createdAt, updatedAt. |
| **TrackAssignment** | userId, track, effectiveFrom/effectiveTo, assignedBy/service, reason, configuration version. |
| **DimensionDefinition** | dimensionId 1–10, name, definition, rubric links, enabled state; official ten dimensions are fixed in v1. |
| **QuestionItem** | questionId, version, track, dimensionId, difficulty, question type, prompt, answer key/rubric, explanation, language, provenance, validation state, hashes/fingerprints. |
| **QuestionExposure** | userId, questionId/version, attempt/session, servedAt, completedAt, exposure state, similarity decision. |
| **AssessmentSession** | sessionId, userId, purpose, track, blueprint/version, scheduledAt, start/end, state, result summary. |
| **AssessmentAttempt** | attemptId, sessionId, userId, question/version, response/artifact reference, times, hints, criterion scores, total, feedback, confidence, validation status, failure reason. |
| **Task / Challenge** | taskId, title, track, dimension targets, instructions, evidence requirements, deadline, rubric/policy version, state, source blueprint. |

| Entity | Required fields / purpose |
|---|---|
| **TaskEvent** | eventId, taskId, userId, previousState, newState, timestamp, actor/service, reason, evidence references. |
| **MasteryEvidence** | evidenceId, userId, dimensionId, source attempt/task, observed score, evidence quality, confidence, policy version, recordedAt. |
| **XPTransaction** | transactionId, userId, amount, reason code, source event, rule version, idempotency key, recordedAt. |
| **RankEvent** | rankEventId, userId, oldRank, newRank, evidence IDs, gates passed, policy version, effectiveAt. |
| **Achievement** | achievementId, userId, badge/achievement type, evidence references, earnedAt, permanent flag. |
| **Competition / Season** | seasonId, competitionId, track/eligibility, dates, rules version, participant variants, scores, standings snapshots. |
| **AIJob** | jobId, job type, input references, model/prompt version, retry count, status, cost/latency metadata, result references, failure code. |
| **AuditEvent** | eventId, actor/service identity, action, target, timestamp, policy version, before/after references where appropriate. |
| **PolicyConfiguration** | configId, type, version, value set, effective dates, author/service, change reason, audit reference. |
| **ReportSnapshot** | reportId, scope, period, metric definitions, source record references, generation metadata, generatedAt. |

### 12.1 Data Integrity Rules

- Use immutable attempt records. Corrections append an adjudication/correction event and preserve the original result.
- All user-owned records must be scoped by authenticated user ID and organization. Client-side filtering is not an authorization boundary.
- Every score or rank event references its rubric/configuration version and source evidence.
- Question exposure must be recorded before the content is revealed to the employee, to close a duplicate-delivery race.
- Use transactions or equivalent atomic operations for XP awards, rank promotions, and idempotent event processing.
- Data must be encrypted in transit and at rest using the selected platform's approved controls.
- Backups must include relational references needed to reconstruct histories. Restore testing is mandatory.

### 12.2 Local Storage

**Architecture requirement**

Browser localStorage may be used for harmless UI preferences or a standalone prototype only. It MUST NOT be the system of record for the production product because multiple employees, cross-device history, background scheduling, admin analytics, and auditability require centrally persisted data.

## 13. AI Quality, Governance and Safety

### 13.1 Content Quality Gate

- Each generated question must have one defensible answer or a rubric capable of accepting valid alternative solutions.
- Validate factual claims in business scenarios. Fictional scenarios must be clearly marked and must not be mistaken for real company data.
- Reject content that is ambiguous, internally contradictory, unsolvable, excessively difficult for its declared level, or inconsistent with the selected track.

- Check accessibility, reading level, cultural/contextual appropriateness, language consistency, and absence of answer leakage.
- Use holdout test fixtures to evaluate question difficulty, answer validity, similarity detection and grader consistency before deploying prompt/model changes.

### 13.2 AI Scoring Guardrails

- AI graders must return structured, schema-validated output with criterion scores, rationale, evidence, and quality/confidence status.
- Use deterministic scoring for fixed-answer questions and computations.
- Grade an employee response against the stored rubric version, not a newly generated rubric after the response is submitted.
- Do not reward excessive verbosity when concise answers meet the task requirement.
- Do not penalize a response for refusing an unsafe instruction, asking for missing facts, or stating uncertainty when appropriate.
- Where the system cannot validate a result, use an automated second evaluation or validation-pending state; never silently invent a score.

### 13.3 Employee Data and Fair Use

- Collect only data needed for development, scoring, system integrity and approved reporting.
- Do not make personality, medical, protected-trait, or emotional-state inferences from biometric data.
- Provide employees understandable explanations of their scores, task requirements, privacy scope and development recommendations.
- Keep development scores separate from sole-source hiring, termination, salary or disciplinary decisions.
- Support authorized correction/appeal workflows without deleting the original attempt or audit evidence.
- Apply the applicable data-protection and employment requirements for the jurisdictions in which the company operates.

### 13.4 AI Model and Prompt Versioning

The system must record model/provider, model version when available, prompt/template version, rubric version, generation settings, and job result. Model changes require regression evaluation on a fixed validation set before rollout. Keep a rollback path and do not silently rescore historical attempts under new rules.

## 14. External Interfaces and Dependencies

| Interface | Requirements |
|---|---|
| **Employee web/mobile UI** | Responsive interface for task completion, feedback, profile/history, ranks, competitions, accessibility, and low-friction daily sessions. |
| **AI model provider** | Secure server-side access for generation, coaching, and rubric scoring; quotas, timeout handling, cost limits, structured outputs, provider fallback where supported. |
| **Persistent database** | Authenticated central storage, atomic transactions, scoped reads/writes, indexes for histories and analytics, backup and restore. |
| **Background scheduler / queue** | Runs training assignments, retention checks, rank evaluations, competition transitions, and reports independently of browser state. |
| **Authentication / identity** | Secure session management, account lifecycle, role mapping and employee-specific access control. |
| **Notification provider** | Approved channels for task reminders and results; consent/settings and deduplication. |
| **Optional business systems** | CRM, project management, HRIS, task tracker or approved evidence integrations. Integration data must be minimized and provenance stored. |
| **Analytics/export** | CSV/JSON and report export within permissions; no uncontrolled public link to sensitive employee histories. |

### 14.1 Technology-Neutral Requirement

This MSRS specifies behaviours and interfaces rather than mandating a particular stack. A production design decision record must select database, AI provider(s), job queue/scheduler, hosting and monitoring components and map them to the requirements above. The design must explicitly avoid making background work dependent on an open client browser.

## 15. Non-Functional Requirements

| ID | Quality attribute | Requirement |
|---|---|---|
| **NFR-001** | Autonomy | Routine assignment, assessment, coaching, scoring, progression, reporting and reminders run without required daily administrator action. |
| **NFR-002** | Availability | Target 99.5% monthly application availability during pilot; production target to be confirmed with hosting/service budgets. Scheduled jobs must recover after transient faults. |
| **NFR-003** | Performance | 95th-percentile dashboard read target <= 3 seconds under agreed pilot load; ordinary non-AI actions should not wait for long AI calls. |
| **NFR-004** | AI latency | Long-running AI jobs must show status, support safe retries, and never block the UI indefinitely. |
| **NFR-005** | Scalability | Support adding employees, question history, tracks and competition seasons without redesigning the data model. |
| **NFR-006** | Security | Authenticated access, server-side authorization, least privilege, secret management, encrypted transport/storage, and security event logging. |
| **NFR-007** | Privacy | Role-limited personal-data access, data minimization, appropriate retention, export controls and audit history. |
| **NFR-008** | Durability | A confirmed assessment attempt must survive browser closure, refresh, device change and application restart. |
| **NFR-009** | Consistency | XP, rank events, history, and seasonal standings must be idempotent and reconstructable from stored records. |
| **NFR-010** | Question freshness | No per-user repeated exact or substantially equivalent questions across recorded exposure history. |
| **NFR-011** | Accessibility | Core flows support keyboard operation, visible focus, semantic labels, readable contrast, and responsive layouts. |
| **NFR-012** | Usability | A daily practice session should have a clear objective, estimated duration, progress indicator, simple submission, and actionable result. |
| **NFR-013** | Auditability | Scores, policies, promotions, exports, overrides and adjudications can be traced to sources, actors/services, and versions. |
| **NFR-014** | Maintainability | Business rules, question policies, role weight profiles, rank thresholds, and prompt versions are centrally versioned and tested. |
| **NFR-015** | Observability | Metrics/logs cover AI generation/grade errors, latency, job queues, retries, duplicate detection, scoring disagreement, costs and storage failures. |
| **NFR-016** | Portability | Employee history and configuration can be exported in a documented machine-readable format and restored into a compatible environment. |
| **NFR-017** | Localization | Assessment language and locale are explicit; scoring rules must avoid hidden language bias where language fluency is not the target skill. |
| **NFR-018** | Cost control | AI calls have per-job, per-user and organization budgets; bounded retries, caching where appropriate, and cost dashboards are required. |
| **NFR-019** | Fairness | Equivalent challenge forms and explicit rubrics are required for comparable competition; monitor scoring differences by role, language, and task form. |

| ID | Quality attribute | Requirement |
|---|---|---|
| **NFR-020** | Recovery | Define retry/backoff, dead-letter handling, backup schedule, restore objectives, and safe-mode behaviour for failed AI providers. |

# 16. Failure Handling and Recovery

| Failure | Required system response |
|---|---|
| **AI provider unavailable** | Retry with bounded backoff; use approved provider fallback or postpone AI-dependent task. Keep browser UI responsive and record job state. |
| **Question invalid / no correct answer** | Reject item before delivery; generate replacement or choose an unseen validated fallback. Do not charge employee with failure. |
| **Question duplicate detected** | Block delivery, record similarity decision, generate/select a different item, and update the exposure guard. |
| **AI grader disagreement** | Run configured automated adjudication; if still unresolved, mark validation pending and exclude from rank gates until resolved automatically. |
| **Database/network error** | Do not show a confirmed submission until persisted. Allow safe retry with an idempotency key and prevent duplicate attempts. |
| **Scheduler outage** | Recover missed jobs on restart using due time and idempotent event processing; record lateness and prevent duplicate task creation. |
| **Employee misses deadline** | Record expired/failed according to policy; retain history, withhold relevant mastery credit, initiate remediation. Separate policy-approved exemptions and system errors. |
| **Invalid evidence or unverifiable outcome** | Mark evidence rejected or insufficient; explain missing criterion and request a new valid submission or fresh task. |
| **Policy version changed** | Apply only according to effective dates; existing attempts retain their original rubric/configuration version. |
| **Restore / recovery event** | Restore profiles, attempt history, question exposures, XP ledger, rank history, and audit chain; run consistency checks before returning to service. |

# 17. Acceptance Criteria

The release is accepted only when the criteria below are demonstrated in a test environment with seeded data and deterministic AI mocks/fixtures where appropriate.

| ID | Acceptance test |
|---|---|
| **AC-001** | A new employee completes onboarding and receives a baseline profile covering all ten dimensions. |
| **AC-002** | Sales, Technical, and Admin tracks produce distinct scenarios for the same selected dimension. |
| **AC-003** | Technical track default tests contain no coding requirement unless the assigned job profile explicitly requires coding. |
| **AC-004** | Every generated question has a validated answer key or rubric before it is shown. |
| **AC-005** | An exact question previously seen by the employee is blocked across all assessment and practice types. |
| **AC-006** | A semantically near-identical question with changed names/numbers is blocked for the same employee. |
| **AC-007** | When generation fails, the system uses a validated unseen fallback or safely defers the session. |
| **AC-008** | Every submitted attempt is persisted and remains visible after refresh, logout/login, and access from another authorized device. |

| ID | Acceptance test |
|---|---|
| **AC-009** | A new attempt never overwrites the old attempt; failed, expired and system-error states remain distinct. |
| **AC-010** | Objective answers are scored by deterministic rules and are not altered by AI-generated prose. |
| **AC-011** | Subjective scores include criterion-level evidence, rubric version, confidence/validation state, and automated adjudication for configured cases. |
| **AC-012** | A low-confidence or disputed score cannot silently unlock a rank. |
| **AC-013** | The system issues a targeted coaching response after failure and creates a different follow-up question. |
| **AC-014** | The system schedules later retention checks with fresh questions. |
| **AC-015** | XP, mastery, competitive rating, and current rank are stored separately. |
| **AC-016** | High XP alone does not grant a rank when assessment or practical gates fail. |
| **AC-017** | A rank promotion records policy version and evidence IDs and can be reproduced from stored rules. |
| **AC-018** | Season reset leaves permanent employee history, rank events and achievements intact. |
| **AC-019** | An admin can filter and inspect the full authorized employee history and all ten dimension trends. |
| **AC-020** | A learner cannot read or alter another learner's records through API or client-side manipulation. |
| **AC-021** | Scheduled tasks and reports execute while the employee and admin browsers are closed. |
| **AC-022** | Reprocessing the same event does not duplicate an attempt, XP award, promotion, badge, or notification. |
| **AC-023** | The system distinguishes employee task failure from platform-generated invalid attempts. |
| **AC-024** | AI-generated admin reports link material conclusions to records and label hypotheses as hypotheses. |
| **AC-025** | Backups are restored in a test and profiles, attempt records, exposure history, and ledgers reconcile. |
| **AC-026** | A failed AI service, job queue, or database request reaches a bounded and visible recovery state. |
| **AC-027** | Changes to rank thresholds or rubrics are versioned; historical attempts are not silently rescored. |
| **AC-028** | A full end-to-end test validates onboarding -> training -> failure coaching -> fresh retest -> mastery update -> rank eligibility -> admin report. |
| **AC-029** | All leaderboard formulas and eligibility rules are visible and reconstructable from the stored policy version. |
| **AC-030** | No production release describes the scores as validated IQ measurements or as sufficient evidence for automatic employment decisions. |

# 18. Verification and Test Strategy

## 18.1 Test Layers

| Layer | What it verifies |
|---|---|
| **Unit** | Scoring formulae, weighted totals, rank gates, XP ledger idempotency, state transitions, deadlines, exact/semantic duplicate guard, permission predicates. |
| **Question-generation contract** | Schema validity, answer key/rubric presence, dimension mapping, track suitability, difficulty, clarity, no answer leakage, safe content. |
| **Catalogue / track matrix** | Every dimension × Sales/Technical/Admin × supported language and question type represented by generated or fixture-based tests. |
| **Integration** | Database persistence, background jobs, exposure records, coaching hand-off, retest scheduling, promotion transaction and reports. |

| Layer | What it verifies |
|---|---|
| **Security** | Authentication, authorization, cross-user access attempts, service identity scope, exports, secrets and audit records. |
| **Reliability** | Provider timeout, rate limit, malformed AI output, adjudication disagreement, queue retry, database outage and recovery. |
| **UX / accessibility** | Keyboard flow, mobile layout, empty states, clear results, understandable rank conditions and error recovery. |
| **Pilot evaluation** | Equivalent but unseen assessment forms, scorer consistency, question difficulty, learning transfer, user engagement and task outcomes. |

### 18.2 Mandatory Adversarial Cases

- An AI returns an answer key that is wrong or admits multiple conflicting answers.
- A model changes an employee score or returns a promotion instruction in free text.
- A generated “new” question copies prior scenario logic with only names and numbers changed.
- The employee refreshes, loses connectivity, submits twice, or finishes after a deadline.
- Two background workers process the same promotion or XP event simultaneously.
- One track’s assessment is materially easier than another track’s assessment but a report compares scores as if equivalent.
- The selected Technical profile is a non-coding role and a generated task nevertheless requires code.
- A season ends while a task is in progress; no permanent history is erased.
- AI quality review fails repeatedly; the system reaches a bounded safe state rather than an infinite loop.
- A platform outage occurs near task deadline; the task is not recorded as employee failure due to system error.

### 18.3 Pilot Measurement

The pilot should measure, at minimum: successful session completion, time to complete, question rejection/duplicate rate, grader disagreement, invalid-task rate, user return/participation, fresh-form assessment change, retention-check results, task quality, practical outcomes, and AI cost per active employee. Do not optimize solely for streaks or screen time.

## 19. Delivery Plan and Release Gates

| Stage | Deliverables | Exit gate |
|---|---|---|
| **Stage 1 - Rules baseline** | Approved ten-dimension definitions, tracks, score profiles, level gates, task states, AI autonomy rules, uniqueness policy, and data model. | Rules are versioned; open policy decisions are documented. |
| **Stage 2 - Assessment core** | Identity/profile, baseline tests, question service, validation and novelty checks, attempt persistence, deterministic scoring. | All question integrity and attempt-history acceptance tests pass. |
| **Stage 3 - AI coaching** | Skill diagnosis, targeted feedback, fresh remediation, adaptive difficulty and spaced retention. | Failure-to-retest loop passes automated end-to-end tests. |
| **Stage 4 - Mastery and game** | XP ledger, six ranks, promotion gates, achievements, seasons, leaderboards and team missions. | No bypass, duplicate award, or loss of career history in adversarial tests. |
| **Stage 5 - Admin and autonomy** | Admin analytics, role-based access, scheduled jobs, reports, monitoring, export, backup and audit trails. | Background operations work without a browser open; permission and recovery tests pass. |
| **Stage 6 - Pilot** | Controlled rollout, question calibration, scorer consistency, engagement and practical-outcome analysis. | Product owner approves measured pilot results and revised defaults. |

### 19.1 MVP Priorities

The first usable release must contain central identity and storage, all ten dimensions, the three tracks, question uniqueness, complete history, basic AI diagnosis/coaching, the six rank labels with configurable gates, and admin access to results. Advanced season animations, integrations, social features, and elaborate badge collections may follow without compromising core integrity.

## 20. Risks, Assumptions and Open Decisions

### 20.1 Primary Risks

| Risk | Impact | Mitigation |
|---|---|---|
| **AI produces wrong questions or scores** | Unfair training, rank changes and loss of trust. | Deterministic checks, stored rubrics, independent evaluation, invalid-item rejection, versioned tests. |
| **Near-duplicate detection misses semantic repeats** | Question repetition undermines learning and score validity. | Combine exact hashes, normalization, semantic similarity, solution-pattern signals, and per-user exposure records. |
| **Gamification encourages easy-task farming** | XP/leaderboards diverge from actual skill. | Separate XP from mastery; rate-limit low-difficulty rewards; verify practical outcomes. |
| **Cross-track scores appear comparable when they are not** | Misleading personnel analysis. | Role-specific rubrics/weights and explicit comparison limits. |
| **Persistent storage or authorization is weak** | Data loss or exposure of sensitive employee history. | Server-side permissions, immutable attempts, security tests, backups, audit logs. |
| **AI operating costs grow without bounds** | Service instability or unaffordable usage. | Budgets, caching, bounded retries, async workflows, provider routing and cost reporting. |
| **Employees optimize for score instead of learning** | Memorization or gaming rather than transferable skill. | Fresh items, transfer tasks, retention checks and measured real-work outcomes. |
| **Overstated “intelligence” claims** | Scores are misinterpreted as fixed traits. | Clear disclaimer; report demonstrated skills and uncertainty, not a single absolute intelligence label. |

### 20.2 Assumptions

- The platform is initially internal to DREAM TEAM SERVICES and supports the company’s employee/team structure.
- Routine operational autonomy is required after initial setup, including background operation when browsers are closed.
- Central persistent storage and server-side jobs are acceptable and necessary for production.
- The three primary tracks are Sales, Technical, and Admin; each maps to the same official ten dimensions.
- Technical roles do not require coding by default.
- Question uniqueness applies at minimum per employee across all assessments and practice, with globally detectable exact/near-duplicate content also rejected where feasible.
- The initial six-rank labels are fixed; thresholds, weightings, XP values, cadence and challenge difficulty remain configurable.

### 20.3 Decisions Required Before Production

| Decision | Proposed baseline | Owner / status |
|---|---|---|
| **Authentication provider** | Use the company’s approved central identity provider. | To confirm |
| **Database / hosting / scheduler** | Select a persistent, transaction-capable datastore plus durable background queue/scheduler. | Engineering decision |

| Decision | Proposed baseline | Owner / status |
|---|---|---|
| **AI providers and fallback** | Define approved primary/fallback models, budgets, retention and data processing terms. | To confirm |
| **Default language and localization** | Configure supported assessment languages; record language per attempt. | To confirm |
| **Rank thresholds and dimension weights** | Use Appendices A/B as pilot defaults; calibrate using evidence. | Pilot decision |
| **Daily training time / cadence** | Proposed target 10-15 minutes on working days with configurable schedules. | To confirm |
| **Data retention and export** | Retain assessment history across employment lifecycle according to approved policy and applicable law. | Privacy decision |
| **Competition visibility** | Publish approved standings; do not publish detailed failure histories. | To confirm |
| **Employee appeals** | Automated re-evaluation first; optional authorized escalation policy. | To confirm |
| **Pilot cohort and duration** | Proposed 30-day pilot across representative Sales, Technical, and Admin users. | To confirm |

# Appendix A. Proposed Role Weightings

These are configurable v1.0 defaults for a pilot only. They are not scientifically validated and must be calibrated using job requirements, scoring reliability, and observed practical outcomes. Each column totals 100%.

| Dimension | Sales | Technical | Admin |
|---|---|---|---|
| **Cognitive Ability** | 5 | 8 | 7 |
| **Logical Reasoning** | 7 | 14 | 9 |
| **Learning Agility** | 10 | 12 | 10 |
| **Critical Thinking** | 10 | 12 | 13 |
| **Problem-Solving** | 12 | 18 | 12 |
| **Creativity & Innovation** | 8 | 8 | 7 |
| **Decision-Making** | 12 | 8 | 14 |
| **Communication** | 15 | 5 | 9 |
| **Emotional Intelligence** | 10 | 4 | 8 |
| **Execution & Adaptability** | 11 | 11 | 11 |
| **TOTAL** | 100 | 100 | 100 |

## A.1 Interpreting Weights

Weights determine the role-specific summary only. They do not change the requirement to track every dimension separately, and must not replace minimum thresholds for critical skills. A person with a weak critical skill cannot become promotion-eligible solely because strengths elsewhere raise the weighted total.

# Appendix B. Proposed Rank Gates

The values below are initial configurable policy defaults for the pilot, not validated cut scores. A level-up requires every gate for that level. “Fresh assessments” means unseen, equivalent-difficulty forms; practical evidence and retention checks are separate from a test percentage.

| Rank | Proposed threshold | Additional gates | Minimum evidence |
|---|---|---|---|
| **1. BASIC** | Entry / baseline | Onboarding complete; baseline evidence recorded. | No prior-rank gate. |
| **2. INTERMEDIATE** | 70%+ on fresh assessment set | Foundational practical task passed; critical criteria met. | One applied task plus baseline completion. |
| **3. ADVANCED** | 75%+ across two distinct assessments | Role-specific challenge passed; consistent dimension minimums. | At least two fresh assessment forms and applied evidence. |
| **4. PRO** | 80%+ across fresh assessments | Independent challenge quality meets rubric; transfer check passed. | Distinct task scenario and positive transfer evidence. |
| **5. PRO MAX** | 85%+ with retention evidence | Verified improvement in a realistic task; critical dimensions meet higher bars. | Delayed retention check and practical outcome evidence. |
| **6. ULTRA PRO MAX** | 90%+ with sustained mastery | Expert capstone passes; retained skill and verified meaningful impact. | Capstone, fresh assessment, retention and impact evidence. |

## B.1 Promotion Invariants

- No level-up from XP alone.
- Every required gate must pass; an overall score cannot conceal a critical dimension failure.
- All promotion evidence is fresh, persisted, versioned and traceable.
- Promotion is automatic when criteria pass; human approval is not required for the routine development workflow.
- Promotion rules are versioned. A later policy change does not silently rewrite historical rank events.

# Appendix C. Illustrative Challenge Patterns

These examples describe the task style and are not live question text. Live questions must be generated as unique variants and pass the question quality gate.

| Track | Dimension | Illustrative challenge | Evidence / scoring focus |
|---|---|---|---|
| **Sales** | Critical Thinking | A lead report shows many inquiries but few purchases. Identify what the data proves and what is missing. | Facts vs assumptions, additional data requests, causal caution. |
| **Sales** | Decision-Making | Choose which of several prospects to contact first with limited time and uneven information. | Prioritization, expected value, reasoning and trade-offs. |
| **Technical** | Cognitive Ability / Learning Agility | Learn a new CRM or AI tool from a short guide, then complete a different workflow task. | Correct application, learning transfer and efficient tool use; no coding by default. |
| **Technical** | Problem-Solving | A website is slow for some users. Review supplied observations and determine the next diagnostic steps. | Evidence-led diagnosis, safe investigation, root cause vs guess. |
| **Admin** | Execution & Adaptability | Complete a scheduling or reporting task, then adapt after the scope changes. | Deadline handling, change management, quality and explanation. |
| **All tracks** | Communication | Explain a proposed solution to a stakeholder who has different background knowledge. | Clarity, structure, audience fit, listening and response quality. |
| **All tracks** | Creativity & Innovation | Generate distinct solutions under a fixed budget and choose one to test. | Originality, usefulness, constraints, feasibility and testability. |

| Track | Dimension | Illustrative challenge | Evidence / scoring focus |
|---|---|---|---|
| **Team mission** | Problem-Solving / Collaboration | Solve a real company workflow bottleneck with role-specific contributions. | Evidence, integration of ideas, individual contribution and verified outcome. |

# End of Specification

### Product rule to protect

AI manages the development process. Evidence establishes skill. Mastery earns the rank. Every failed attempt becomes a targeted learning opportunity. The record of what happened is never silently overwritten.